



QP CODE: 23709215



Reg No :

Name :

M.C.A. DEGREE EXAMINATION, MAY 2023

First Semester

Master of Computer Application

Core - MCACT105 - DATABASE TECHNOLOGY AND NOSQL

2020 Admission Onwards

98CE9EF0

Time: 3 Hours

Maximum: 75 Marks

Part A

*Answer any **ten** questions*

*Each question carries **3** marks*

1. Discuss the disadvantages of DBMS.
2. What is a data model?
3. Explain the different types of attributes.
4. Compare Primary key and foreign key with suitable examples?
5. Discuss the rules used to convert ER diagram to tables.
6. Represent weak entity sets through a suitable example and explain its operations.
7. What is index ? Explain with Example
8. What are basic set operations in SQL?
9. Compare Serial Schedule with Concurrent Schedule.
10. Compare Commit and Roll Back Commands.
11. Describe Aggregation with suitable examples.
12. Explain the difference between sharding and replication.

(10×3=30 marks)





Part B

Answer **all** questions

Each question carries **9** marks

13. a) Demonstrate the advantages and disadvantages of using a database management system.

OR

- b) Explain how a database can be designed for a Student Information System.

14. a) Briefly write a note on Integrity Constraints.

OR

- b) Define the procedures for transforming ER model to relational model with suitable diagrams.

15. a) Explain Embedded SQL, Dynamic SQL and Cursors with example.

OR

- b) Give a set of FDs for the relation schema $R(A, B, C, D)$ with primary key AB under which R is in 2NF but not in 3NF. Justify your answer.

16. a) Explain 2 Phase Locking with its variants.

OR

- b) Explain how we can Design Distributed DBMS.

17. a) Illustrate types of NoSql databases with suitable example.

OR

- b) Discuss Distribution models in NoSQL.

(5×9=45 marks)

